

- Time: Summer 2026, Tuesdays 14:30 pm to 17:45 pm
- Location: Augustenstrasse 40, F.1.11, Seminarraum (2913.01.111)
Instructors: Shaghayegh Kolli (shaghayegh.kolli@tum.de), Jana Diesner (jana.diesner@tum.de), Nafiseh Nikeghbal (Nafiseh.Nikeghbal@tum.de)
- Course language: English
- Workload: 10 ECTS

1. Description

This module brings together practical approaches from the fields of human-centered data science, which considers societal contexts and ethical concerns for data analysis, and computational social science, which leverages large-scale data to discover theories and patterns of societal behavior. Work in these areas matters as solving the hard and pressing problem of our and future generations requires teams that combine the how (data, algorithms, tools) with the why (societal implications, culture, ethics) of computing with human-centered data and/ or applications that impact people's lives. We will use concepts, theories, methods, algorithms, and evaluation methods from the fields/ domains of natural language processing, network science, statistics, machine learning, AI, and the social sciences.

This module entails: 1) implementing data analysis and/ or software (sub)project in teams, 2) practicing methods and models of computational data analysis and/ or software development and/ or programming from the outlined areas, and 3) practicing techniques for planning, documentation, assessing, and presenting process and (intermediate) results.

2. Pre- and Co-requisites

Skills and knowledge in/ of deep learning, machine learning, and natural language processing.

3. Student Learning Outcomes

Completing this module and its requirements should enable students to acquire the following knowledge and skills:

- Conceptualize, plan, design, implement, and evaluate a computational research or data analysis project, tailored to topics such as language modeling, misinformation detection, or literary network analysis — all within a team-based research environment.
- Integrate theories, tools, and methods from natural language processing (NLP), computational linguistics, AI, and digital humanities.
- Identify and address technical and methodological challenges such as bias in AI models, alignment with human values, multimodal input processing, or ambiguity in language and annotation — and troubleshoot them for practical/ downstream applications.
- Understand and reflect on the ethical, cultural, and contextual dimensions of AI applications, e.g., when dealing with misinformation, human value alignment, multilingual content, or narrative representation.

- Communicate their research process and outcomes effectively — both in written form (research reports, annotated datasets) and through visual or oral formats (e.g., poster presentations, system demos).
- Strengthening academic and professional communication skills, including collaboration within interdisciplinary teams, iterative project development, and presenting to both technical and non-technical audiences.

4. Teaching and Learning Methods

Participants practice systematic computational data analysis and programming techniques in teams with a given task description and deliverables (design, implementation, evaluation, interpretation, documentation, dissemination). Intermediate results of the teamwork will be presented. Students participate in exercises in the classroom and at home. Although we meet in person, some content will be delivered online.

5. Week-by-week topic and assignment schedule

The schedule and readings are subject to change. The most up to date syllabus and set of readings and deliverables are available on the course Moodle.

Week	Date	Theme	Deliverables
1	Tue, 14 April 2025	Self-study Package	
2	Tue, 21 April 2026	Meet & greet; Module overview; Team formation	
2	Tue, 28 April 2026	Literature review workshop; Related work & tools	Student progress presentation
3	Tue, 5 May 2026	Research questions; Project plan	Student progress presentation
4	Tue, 12 May 2026	Research design & methodology	Student progress presentation
5	Tue, 19 May 2026	Dataset selection & scoping	Student progress presentation
6	Tue, 26 May 2026	Data pipeline & preprocessing I	Student progress presentation
7	Tue, 2 Jun 2026	Data pipeline & preprocessing II	Student progress presentation
8	Tue, 09 Jun 2026	Implementation phase I	Student progress presentation
9	Tue, 16 Jun 2026	Implementation phase II	Student progress presentation
10	Tue, 23 Jun 2026	Implementation phase III	Student progress presentation
11	Tue, 30 Jun 2026	Metrics & troubleshooting	Student progress presentation
12	Tue, 07 July 2026	Model improvements + error analysis	Student progress presentation
13	Tue, 17 July 2026	Model improvements + error analysis	Student progress presentation

16	Tue, 21 July 2026	Poster presentation day	Poster presentation
17	Tue, 31 July 2026	Report submission; wrap-up	Report submission

6. Course materials

Readings (required and optional) will be announced on Moodle as needed.

7. Assignments, Methods of Assessment, Expectations related to Deliverables

This module is designed to be highly interactive and practice-oriented. Students will engage in continuous learning through weekly presentations, discussions, project work, and final deliverables. The purpose of these assignments is to enhance students' ability to conceptualize, analyze, and present computational data projects while fostering teamwork, critical thinking, and academic writing skills.

• Module Requirements:

- Assignments:** Students will complete various assignments, including project-related tasks, research paper presentations, and discussions. Assignments are structured to build progressively towards the final project presentation and paper report.
- Presentation:** you will present your project weekly.
- Project related assignments:** Students will work in small teams on a project. Throughout the course, teams will present their project progress, receive feedback, and refine their work. This will culminate in a final presentation and written report detailing their findings.
- Readings and Discussions:** Students are expected to complete assigned readings posted on the course website and actively participate in discussions. These readings provide foundational knowledge necessary for successful project completion and engagement with course material.

• Evaluation and grading policy

Deliverable	Grade
- Weekly presentations	30%
- Final project presentation	35%
- Final report	35%

• Grading Scale

100 - 95	1	very good
94 - 90	1.3	very good
89 - 85	1.7	good
84 - 80	2	good
79 - 75	2.3	good
74 - 70	2.7	satisfactory
69 - 65	3	satisfactory
64 - 60	3.3	satisfactory

59 - 55	3.7	passed
54 - 50	4	passed
less		failed/ not passed

- **Additional Expectations**

- All work must be completed to pass this module.
- If you encounter an emergency that might affect your ability to meet deadlines or contribute to group work, please communicate with the lead instructor (Sha) as early as possible. Timely communication can help prevent detrimental impacts on your grade or your group's progress.
- Since this is a group-based practical course, collaboration within your assigned group is expected and encouraged. All group members are responsible for contributing meaningfully to the project and for understanding the work submitted under their names.
- You are allowed to use and adapt the work of others (eg, pre-trained models, open-source datasets, existing codebases), if the following conditions are met:
- You do not violate copyright or licensing agreements.
- You clearly attribute any reused or adapted content, including code, data, diagrams, or text.
- You document what changes or additions your team made, and what portion is original.
- All written submissions (e.g., project reports, summaries, code comments) must be the original work of your group. If your writing includes ideas or quotes from others (including research papers or documentation), you must cite them properly.
- Academic dishonesty, including plagiarism or misrepresentation of work, will not be tolerated. Violations will be handled according to university policy and may result in a reduced grade, failure of the assignment or course, or disciplinary action.

8. Course Policies

- **Attendance/ Participation Policy**

- We expect students to 1) attend all sessions except in cases of emergency and 2) to show up on time.
- Enrollment in this course includes the expectation of regular attendance and being present when we start. If you find you must miss (or have missed) a module or need to be late, please inform the lead instructor.
- **Exception/ accommodation:** You may miss one session without an excuse or explanation.

- **Module conduct:**

- You share some of the responsibility for fostering an inclusive classroom. You are expected to be respectful of others' perspectives and lived experiences during discussions.
- You are expected to demonstrate respect for the ideas and opinions of all other participants in this module at all times. Failure to observe this requirement can result in a failing grade for the module.

- **Academic Integrity**

You are asked to adhere to TUM's Code of Conduct https://gc.gs.tum.de/wp-content/uploads/2021/08/TUM_RiLi_GWP_2015_eng.pdf and the Code of Academic Practice at https://www.gs.tum.de/fileadmin/w00bik/www/pdf/TUM_SGwP_en.pdf.

9. Additional Information

• Statement of Inclusion

TUM values the diversity and variety of individual talents as a central element of its institutional identity. You are asked to adhere to TUM's Diversity Code of Conduct (<https://www.zv.tum.de/en/diversity/inclusion/tum-diversity-code-of-conduct/>).

• Religious Observances

I aim to reasonably accommodate the participants' religious beliefs, observances, and practices. If you anticipate the need for accommodation, please communicate with me in the first two weeks of module.

• Accessibility Statement

If you need accessibility-related academic adjustments and/or auxiliary aids, please contact the Departmental Contact for Disabled and Chronically Ill Students as soon as possible, who will advise on next steps. You can also contact the campus-level unit for "Studying with Special Needs" at Handicap@zv.tum.de. We also aim to adhere to TUM's accessibility statement (<https://www.tum.de/en/spezielseiten/accessibility>).

To ensure that disability-related concerns are properly addressed from the beginning of the semester, I request that students with disabilities who require assistance to participate in this module contact me as soon as possible to discuss their needs and any concerns they may have. I do not require a letter from the unit for "Studying with Special Needs" to discuss your requested accommodations.

• Resources to help you succeed:

- Library Resources: <https://www.ub.tum.de/>.
- Writing Assistance: TUM offers various programs to improve writing skills in English and German for all TUM members. <https://www.sprachenzentrum.tum.de/en/sz/writing-assistance/>.

10. Bio of Instructors

Shaghayegh Kolli is a doctoral student at the Technical University of Munich with a background in Computer Engineering and Media Management. Her research focuses on Natural Language Processing, Machine Learning, and Generative AI. She works on projects like fake news detection, multilingual text generation, and AI-human interaction.

Nafiseh Nikeghbal is a doctoral student at the Technical University of Munich, in the Human-Centered Computing Group. Her work involves projects on the behavior, structure, and potential of LLMs, with a focus on their capabilities and social biases.

Jana Diesner is a Full Professor for Human Centered Computing at the Technical University of Munich. Prior to that, she was a tenured Associate Professor at the School of Information Sciences at the University of Illinois Urbana-Champaign. Dr. Diesner's basic and applied research in

human-centered data science and responsible computing combines the benefits of machine learning, AI, network analysis and natural language processing with the consideration of social science theories, contexts and culture, and ethical concerns. Jana received her PhD from Carnegie Mellon University's School of Computer Science.

11. Details on team work

Teams: You are welcome to work in teams of 2-5 people.

Process: You will be introduced to research process early in module. The deliverables for the project are aligned with the steps of that process.

Data: You can analyze existing data or generate data. Collecting data as part of this project might be very time consuming and require ethics considerations. You are asked to consult with me if they have questions about this point.

Report: You write up your work and findings in the form of a research report. You will receive feedback on each deliverable. You need to structure your report as specified below. If you plan to submit your manuscript for review for publication you can use the style sheet provided or required by that venue.

Project report structure and evaluation

1. Title
2. Name(s) of author(s).
3. Keywords (self-defined)
4. Abstract
 - a. 150 to 250 words.
 - b. Motivate and state your research question.
 - c. Focus on what you have learned (key findings).
5. Introduction
 - a. Why does your research question matter?
 - b. Develop your research question or research problem.
6. Background
 - a. Discuss and synthesize prior work such that you identify a gap in prior research, a conflict between prior results, or a lack of knowledge with respect to understanding the problems centric to this course. Start by drawing from the course readings and expand with additional readings.
7. Data
 - a. Describe the dataset you use. This description should include:
 - i. How the data was collected (by who, from whom, when, what methods).
 - ii. Size.
 - iii. Reliability.
 - iv. Limitations.
8. Method
 - a. How do you analyze the data (unit of analysis, metrics, etc.).
9. Results

- a. Describe your findings. Add in figures, tables and visualizations as needed.
- 10. Conclusions
 - a. Interpret your findings with respect to your research question.
 - b. What new knowledge was gained from this study?
 - c. How can your findings be expected to generalize?
 - d. For whom might your findings be relevant, who can use the knowledge you have gained?
- 11. Limitations and Future Work
 - a. State all limitations that apply to the data, methods and results.
 - b. How could your work be extended and why would that be useful?
 - c. Any ethical considerations?
- 12. References
- 13. Appendix